MORPHISMS BETWEEN FORMAL SPECTRA COME FROM CONTINUOUS RING MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms from an affine scheme into a formal spectrum factor through a quotient, and the universal property of the spectrum of a ring. Same chapter, Lemma lemma-factor-through-thickening at proof lines 2364-2367, and Schemes, Lemma schemes-lemma-morphism-into-affine at proof lines 2367-2369. Produce the compatible algebra maps between the quotients.
- Weakly admissible topological rings and their weak ideals of definition. Same chapter, the definitions in force in the statement. Fix the systems of ideals over which the construction is made.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01I1: sources/stacks-chow/tags.txt, source lines 1962–1962. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let A, B be weakly admissible topological rings over S. Any morphism $f : Spf(B) \to Spf(A)$ of affine formal algebraic spaces over S is equal to $Spf(f^\sharp)$ for a unique continuous S-algebra map $f^\sharp : A \to B$.*

Proof. Let $f : \mathrm{Spf}(B) \to \mathrm{Spf}(A)$ be as in the lemma. Let $J \subset B$ be a weak ideal of definition. By Lemma 0AIA there exists a weak ideal of definition $I \subset A$ such that $\mathrm{Spec}(B/J) \to \mathrm{Spf}(B) \to \mathrm{Spf}(A)$ factors through $\mathrm{Spec}(A/I)$. By Schemes, Lemma 01I1 we obtain an S-algebra map $A/I \to B/J$. These maps are compatible for varying J and define the map $f^\sharp : A \to B$. This map is continuous because for every weak ideal of definition $J \subset B$ there is a weak ideal of definition $I \subset A$ such

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

that $f^\sharp(I) \subset J$. The equality $f = \mathrm{Spf}(f^\sharp)$ holds by our choice of the ring maps $A/I \to B/J$ which make up $f^\sharp$. $\square$